

Yao-Rui Yeo

CONTACT INFORMATION

New York University
180 Madison Ave, 17th Floor
New York, NY 10016

Website: <https://yaoruiy9955.github.io/>
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EDUCATION

Ph.D. in Mathematics, University of Pennsylvania 2016 Aug – 2021 May
Thesis advisor: Ching-Li Chai
Thesis title: Mixed Carlitz motives and colored multizeta values in characteristic p
B.A. in Mathematics, University of Minnesota 2013 Aug – 2016 May
Honor: Summa cum laude

EMPLOYMENT

Research Scientist, New York University 2023 Feb – Present
Senior Systems Analyst, New York University 2021 Aug – 2023 Feb

RESEARCH INTERESTS

Health Economics and Disease Modeling
Health equity; Inequality indices; Population health metrics; Decision analytic modeling
Algebraic Number Theory and Combinatorics
Arithmetic and geometry of zeta functions; Special functions, graphs, and their applications

PAPERS

(* = 1ST/CO-1ST)

1. Cebisile Ngcamphalala, **Yao-Rui Yeo**, Mellesia Jeetoo, Dyanna Charles, Zwakele Tfwala, Nokuthula Mahlalela, Lenhle Dube, Ruben Sahabo, Mfundu Motsa, Harriet Nuwagaba-Biribonwoha, Scott Braithwaite, *Cost-effectiveness of implementing long acting PrEP in Eswatini: A mathematical modeling study*, Lancet Reg. Health Afr., to appear.
2. Cebisile Ngcamphalala, **Yao-Rui Yeo**, Jasmine Su, Mellesia Jeetoo, Dyanna Charles, Zwakele Tfwala, Nokuthula Mahlalela, Lenhle Dube, Ruben Sahabo, Tengetile Dlamini, Harriet Nuwagaba-Biribonwoha, R. Scott Braithwaite, *Determining minimum resources required by Eswatini to meet its Ending the HIV Epidemic 2030 goal*, JAIDS **101** (2026), Issue 4, 390–398.
3. Tsitsi Apollo, **Yao-Rui Yeo**, Owen Mugurungi, Ngwarai Sithole, Isaac Taramusi, Emmanuel Tachiwenyika, Emily Gwavava, Wanzirai Majoni, Anesu Chimwaza, Kudawashe Takarinda, Getrude Ncube, Mellesia Jeetoo, Dyanna Charles, R. Scott Braithwaite. *Case study of PEPFAR funding cuts on HIV infections and deaths: Reversing hard-won gains*, AIDS **39** (2025), Issue 14, 2143–2146.
4. Jasmine Su, **Yao-Rui Yeo**, Mellesia Jeetoo, Neo Morojele, Joel Francis, Sheela Shenoi, R. Scott Braithwaite, *Cost-effectiveness of screening and treating alcohol use and depression among people living with HIV in Zimbabwe: a mathematical modelling study*. BMC Med. **22** (2024), No. 481.
5. **Yao-Rui Yeo***, *Algebraic independence results for colored multizeta values in characteristic p* , Int. J. Number Theory **19** (2023), No. 3, 677–708.
6. Logan Crew*, Alexander Kirillov*, **Yao-Rui Yeo***, *Branching rules for splint root systems*, Algebr. Representat. Theory **25** (2022), 963–981.
7. Wan-Jin Yeo*, **Yao-Rui Yeo***, Samu Taulu, *Towards analytical calculation of the magnetic flux measured by magnetometers*, Phys. Lett. A **411** (2021), 127545.
8. Yao-Yu Yeo*, **Yao-Rui Yeo***, Wan-Jin Yeo*, *A computational model for estimating the progression of coronavirus cases in the US West and East Coasts*, Exp. Results **1** (2020), e41.
9. Wan-Jin Yeo*, **Yao-Rui Yeo***, *Enumerating (k, l) -critical and supercritical permutations*, Math. Medley **41** (2015), No. 1.

PREPRINTS

1. **Yao-Rui Yeo***, *Regular intersection graphs of line segments*.
2. **Yao-Rui Yeo***, *Equally-distributed-equivalent from an L -function viewpoint*.
3. **Yao-Rui Yeo***, *Applying Atkinson's inequality aversion index to health metrics*.
4. **Yao-Rui Yeo***, Mellesia Jeetoo, Dyanna Charles, Cebisile Ngcamphalala, Sheela Shenoi, R. Scott Braithwaite, *Could using WALYs rather than QALYs influence inferences from allocative efficiency results? A case study of HIV-related interventions in Southern Africa*.
5. **Yao-Rui Yeo***, Mellesia Jeetoo, Dyanna Charles, R. Scott Braithwaite, *Cost-effectiveness of implementing long acting antiretroviral therapy in Zimbabwe: A modeling study*.

	6. Daniel Citron*, Yao-Rui Yeo* , Cebisile Ngcamphalala*, Kasturi Bhamidipati, Harriet Nuwagaba-Biribonwoha, Zwakele Tfwala, Samkelo Simelane, Trevor Sithole, Mfundi Motsa, Sindi Matse, Velephi Okello, Anna Bershteyn, R. Scott Braithwaite, <i>Health impact of bilateral aid disruptions to HIV services in Eswatini: insights from a two-model study</i> .
INVITED PRESENTATIONS	1. The 26th International AIDS Conference, <i>Cost-effectiveness of online pharmacy-delivered PrEP in California: The PrEP-3D RCT</i>, Poster Presentation, Rio de Janeiro, 2026 Jul 28. 2. Society for Medical Decision Making 48th Annual Meeting, <i>Cost-effectiveness of implementing long-acting antiretroviral therapy in Zimbabwe: A modeling study</i>, Poster Presentation, Oslo, 2026 Jun 30. 3. Society for Medical Decision Making 48th Annual Meeting, <i>A new mathematical formulation of inequality aversion in health</i>, Poster Presentation, Oslo, 2026 Jun 28. 4. Conference on Retroviruses and Opportunistic Infections, <i>Modeled comparison of well-being and quality-adjusted-life-years as HIV outcomes in Southern Africa</i>, Poster Presentation, Denver, 2026 Feb 24. 5. Joint Mathematics Meetings, <i>Regular intersection graphs of line segments</i>, Oral presentation, Washington DC, 2026 Jan 07. 6. Society for Medical Decision Making 47th Annual Meeting, <i>How much do allocative efficiency analyses change by using well-being-adjusted life-years rather than quality-adjusted life-years?</i>, Oral Presentation, Ann Arbor, 2025 Jun 16. 7. Conference on Retroviruses and Opportunistic Infections, <i>Cost-effectiveness of alternative targeting strategies for 3 types of PrEP in Zimbabwe</i>, Poster Presentation, San Francisco, 2025 Mar 11. 8. University of Pennsylvania, Algebra Seminar, <i>Colored multizeta values in characteristic p</i>, Oral Presentation, Philadelphia, 2021 Mar 19. 9. Temple University, Fifth Annual Graduate Student Conference, <i>Branching rules for splint root systems</i>, Oral Presentation, Philadelphia, 2019 Jun 01. 10. University of Minnesota, Undergraduate Honors Seminar, <i>Dessins d'enfants and superpotentials</i>, Oral Presentation, Minneapolis, 2016 May 03. 11. University of Illinois at Chicago, Undergraduate Mathematics Symposium, <i>The Hurwitz existence problem</i>, Oral Presentation, Chicago, 2015 Oct 24.
STAKEHOLDER PRESENTATIONS	1. Shelby County Health Department, <i>Allocative efficiency modeling in Memphis, 3rd workshop</i>, Onsite Presentation, 2025 Nov 13. 2. Shelby County Health Department, <i>Allocative Efficiency Modeling in Memphis, 2nd workshop</i>, Onsite presentation, 2025 May 22. 3. Shelby County Health Department, <i>Introduction to HIV intervention prioritization in Memphis</i>, Onsite Presentation, 2025 Feb 13. 4. Eswatini Ministry of Health, <i>Allocative efficiency modeling for Eswatini to meet EHE 2030 goals</i>, Online Presentation, 2023 May 10. 5. Zimbabwe Ministry of Health and Child Care, <i>Allocative efficiency modeling for Zimbabwe to meet EHE 2030 goals</i>, Online Presentation, 2023 Mar 29. 6. KwaZulu-Natal Department of Health, <i>Allocative efficiency modeling for Kwazuku-Natal to meet EHE 2030 goals</i>, Online Presentation, 2023 Mar 21. 7. New York City Department of Health and Mental Hygiene, <i>Rapid COVID-19 modeling on Delta and Omicron variants</i>, Bi-weekly Online Presentations, 2021 Aug – 2022 Dec.
INVITED LECTURES	1. University of Washington, Institute for Learning and Brain Sciences, <i>Analytical and numerical methods for special differential equations</i>, 5-Hour Lecture, 2019 Dec – 2020 Jan.
OTHER INVITES	1. New York University, <i>Poster Judge</i>, Trainee Research Day, 2025 Apr 22. 2. University of Pennsylvania, <i>Booth Designer</i>, Math Fest, 2018 – 2020.
AWARDS	Distinguished Service Award , New York City Department of Health 2021
TEACHING	Tutoring Service of New York 2021 – 2025 <i>Subjects: Mathematics, Statistics, Physics.</i>

Directed Reading Program, University of Pennsylvania	<i>2020 – 2021</i>
<i>Alex Kalbach, Spring 2021. Topic: The Fifteen Theorem.</i>	
<i>Mihal Zelenin, Fall 2020. Topic: Traffic Flow Modeling and Car Accident Risk.</i>	
<i>Esther Guan, Spring 2020. Topic: Cryptosystems.</i>	
Graduate Student Instructor, University of Pennsylvania	<i>2019</i>
<i>MATH 240 (Calculus III), Summer 2019.</i>	
Teaching Assistant, University of Pennsylvania	<i>2017 – 2021</i>
<i>MATH 603 (Algebra), Spring 2021.</i>	
<i>AMCS 602 (Applied Algebra), Fall 2020.</i>	
<i>MATH 241 (Calculus IV), Spring 2019.</i>	
<i>MATH 420 (Ordinary Differential Equations), Fall 2018.</i>	
<i>MATH 240 (Calculus III), Spring 2018.</i>	
<i>MATH 371 (Algebra), Fall 2017.</i>	
<i>MATH 110 (Calculus for Wharton Students), Fall 2017.</i>	
<i>High School Summer Mathematics Academy, Summer 2017.</i>	

OTHERS

Citizenship: Taiwan
Permanent Residence: USA
Music: Fellow of Trinity College London in Piano Performance